When Payoff Returns: Reinstatement of Earlier Narrative Context During Recall

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Takeaway

- During viewing, plant-payoff pairs were **not detectable** as sustained pattern similarity.
- When participants recall a payoff later, posterior medial cortex (PMC) patterns resemble the pattern of **its own plant** more than other plants.

Background

Does the brain hold on to the link between **a plant and its payoff?**

- Concept from screenwriting
- Usually **don't function as boundaries**
- Usually **across events**





The stirred teacup that later triggers hypnosis. Picture from *Get Out* (Peele, 2017) © Universal Pictures.

Materials & Methods

- Data: Sherlock (Chen et al., 2017). **N = 17**; Sherlock S1E1, **Viewing + Free Recall, fMRI**
- **Pre-specified** annotated plant-payoff pairs, **N = 11** (modality, timing, cue, characters)
- Identity percentile: rank of the correct plant among all earlier scenes in PMC pattern similarity to the recalled payoff (0.5 = chance).
- One annotator

Research Questions

- As meaning changes, does the brain encode the plant and the payoff in **similar patterns**?
- How is the plant-payoff relationship expressed when the payoff is recalled?

"If in the first act you have hung a pistol on the wall, then in the following one it should be fired." — Anton Chekhov

Results

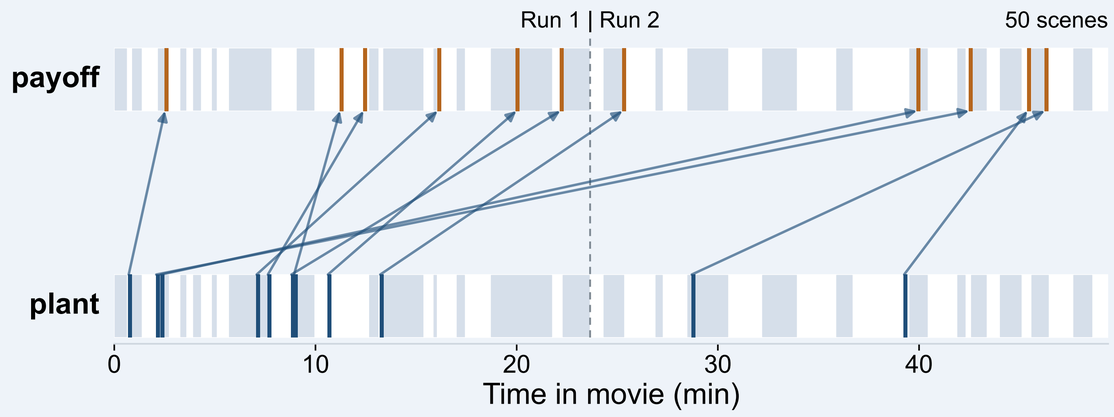


Figure 1. Annotated plant-payoff pairs across the material.
The episode was presented in 2 scanning runs, each preceded by the same 30 second cartoon (excluded from the analysis).
Shading shows the 50 scenes of Chen et al. (2017). Each arrow represents one of the 11 locked pairs, from the plant (bottom) to the payoff (top).

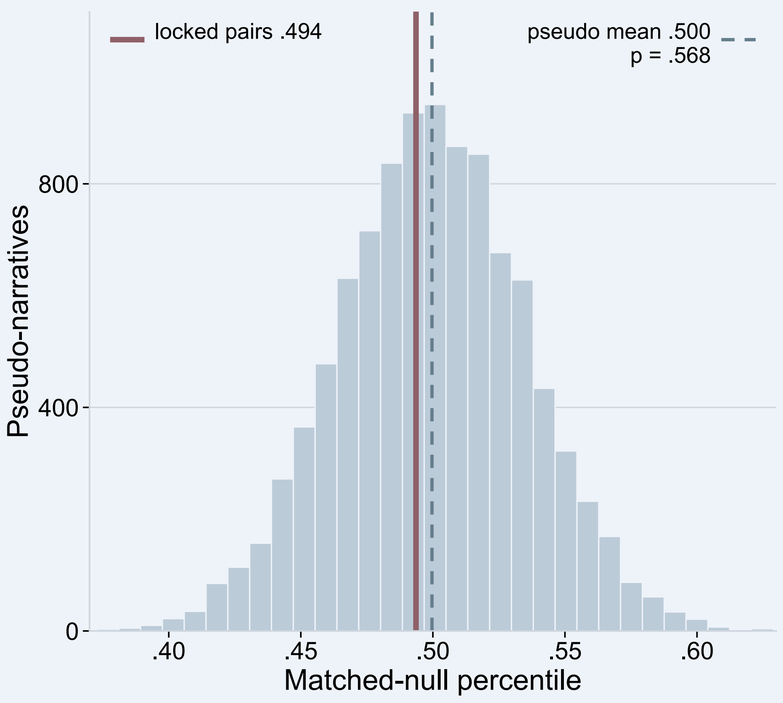


Figure 2. Plant-payoff structure was undetectable during viewing.
Plant-payoff similarity was compared with control pairs matched on temporal gap. Same result was shown at the exact-gap sensitivity analysis (0.534, p = 0.148).

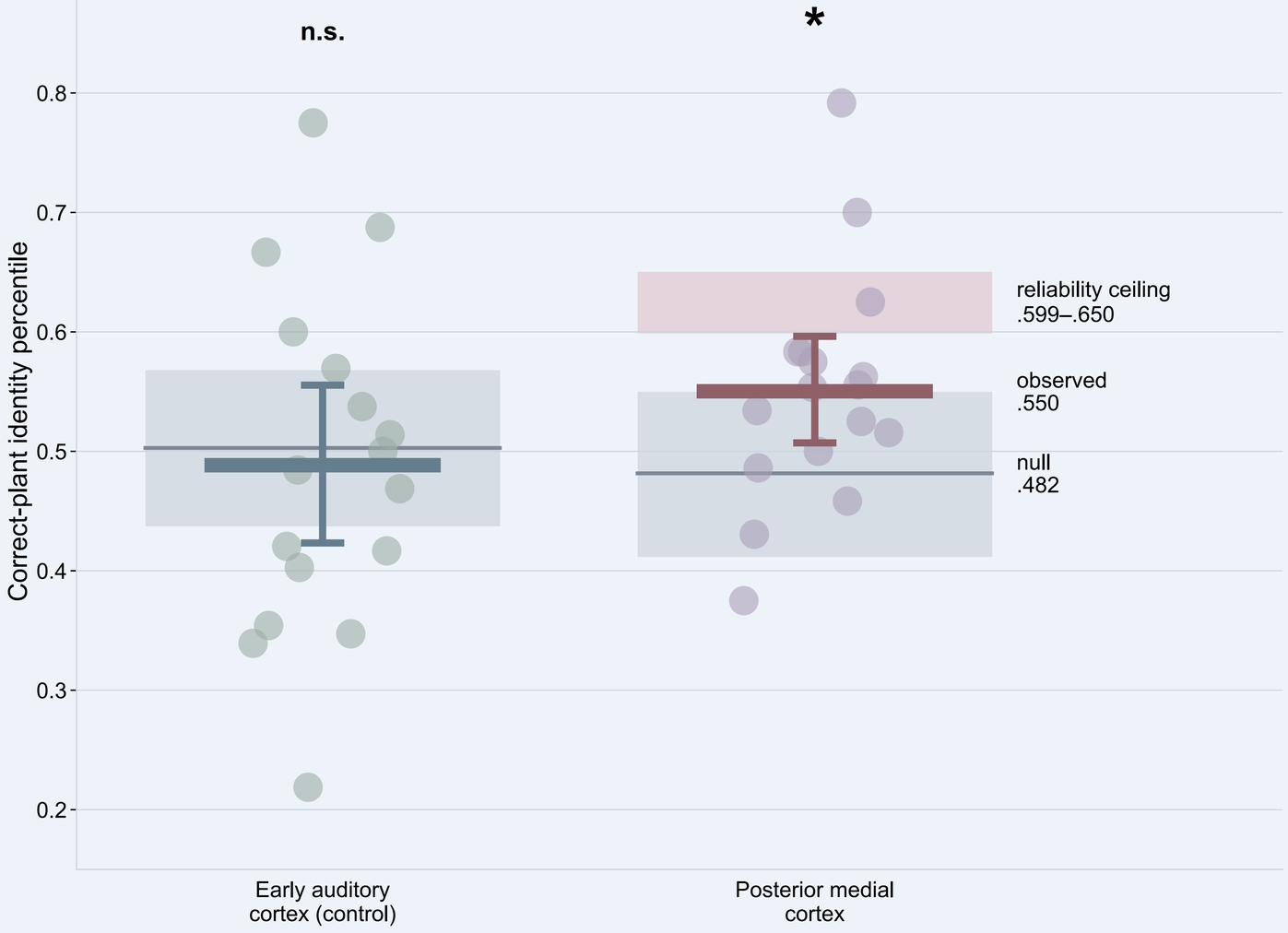


Figure 3. In PMC, recalled payoffs resembled their own plant more than other plants.
Points: participant-level identity percentiles; thick lines and error bars: group mean and 95% bootstrap CI. Grey: order-valid null; rose: reliability ceiling. q < 0.05, FDR across ROIs.

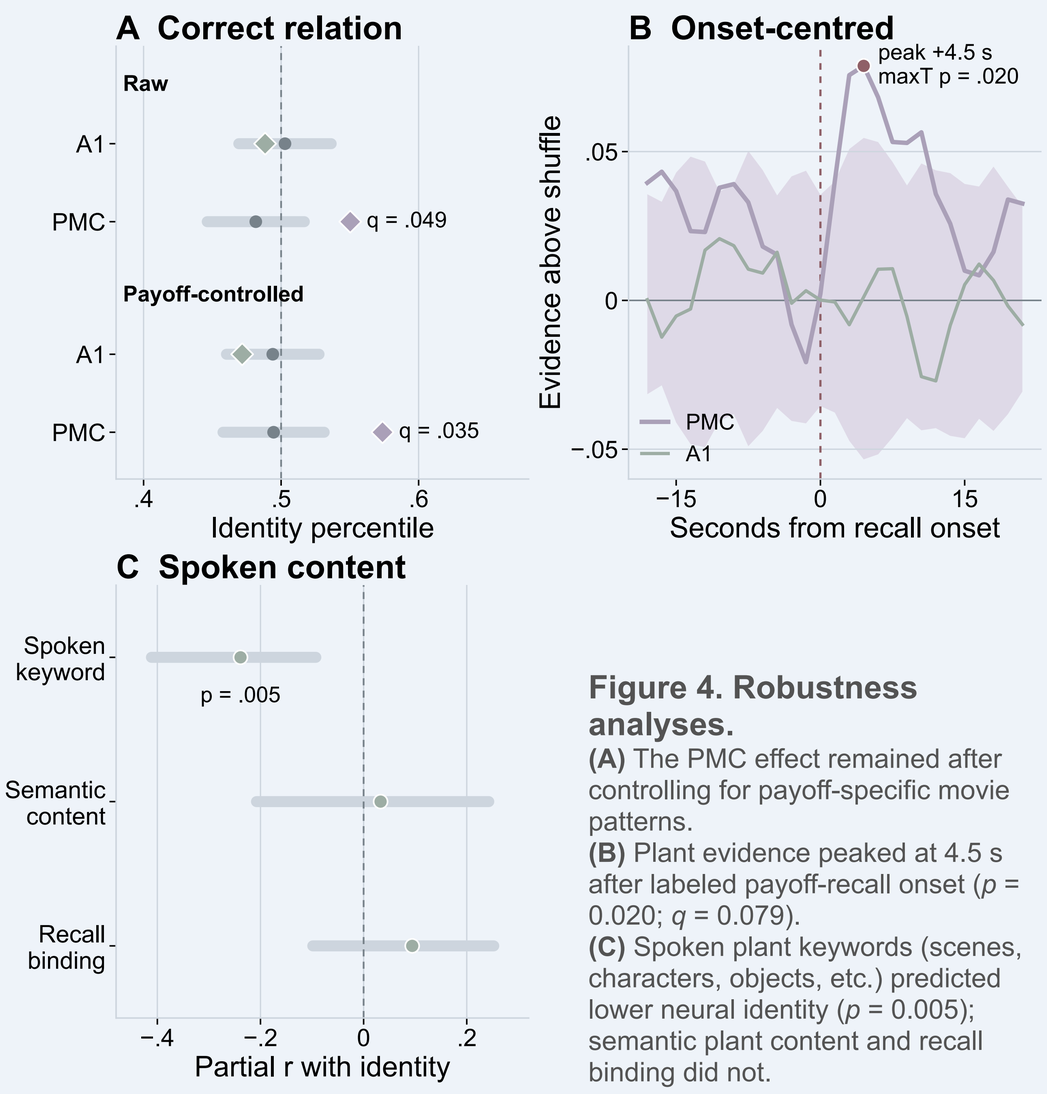


Figure 4. Robustness analyses.
(A) The PMC effect remained after controlling for payoff-specific movie patterns.
(B) Plant evidence peaked at 4.5 s after labeled payoff-recall onset (p = 0.020; q = 0.079).
(C) Spoken plant keywords (scenes, characters, objects, etc.) predicted lower neural identity (p = 0.005); semantic plant content and recall binding did not.

Null: 10,000 relabelings preserving plant-before-payoff order.
Sensitivity: LOSO p = 0.005 - 0.037; LOPO (pair) p = 0.004 - 0.188 (5/11 ≥ 0.05).
Reliability ceiling: score of the earlier scene ranked most consistently across participants (LOSO-to-all-participant bound).

Discussion

Viewing

No stable plant-payoff similarity within this design's sensitivity.

Recall

Recalling a payoff brings back its plant's context in PMC, around recall onset.

Next steps

Across datasets
Number of annotators
Other storytelling techniques

Conclusion

Memory reconstruction, rather than continuous representation, may reveal long-range narrative links.

Looking for a PhD position in event cognition and naturalistic memory
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References
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Baldassano C, et al. Discovering event structure in continuous narrative perception and memory. Neuron. 2017;95:709–721.